

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks available |     |     |       |       |      |
|----------|-----|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4        | (a) |  |  | <div>the <i>s</i>-block, <i>p</i>-block and <i>d</i>-block <input checked="" type="checkbox"/></div> <div>the <i>s</i>-block and <i>p</i>-block only <input type="checkbox"/></div> <div>the <i>s</i>-block only <input type="checkbox"/></div> <div>the <i>d</i>-block and <i>p</i>-block only <input type="checkbox"/></div> <div>the <i>p</i>-block only <input type="checkbox"/></div>                                                                                                                                                                                                                                                                                                              | 1               |     |     | 1     |       |      |
|          | (b) |  |  | titanium / Ti                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 |     | 1   | 1     | 1     |      |
|          | (c) |  |  | <div>the Group 1 metals and transition metals all have a +1 oxidation state <input type="checkbox"/></div> <div>the transition metals all have a +3 oxidation state <input checked="" type="checkbox"/></div> <div>the Group 1 metals all have a +1 oxidation state <input checked="" type="checkbox"/></div> <div>iron and lithium have the same oxidation states <input type="checkbox"/></div> <div>the Group 1 metals and transition metals all have a +4 oxidation state <input type="checkbox"/></div> <div>award (1) for each correct tick</div> <div>if more than two ticks</div> <div>award (1) for two correct and one incorrect</div> <div>award (0) for one correct and two incorrect</div> |                 |     | 2   | 2     |       |      |